SDS8102: Advanced Mathematical Statistics

Lecture 2: Statistics, Sufficiency, Completeness, and Decision Theory

Saptarshi Roy

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Department of Statistics and Data Science
MBZUAI

Statistics and Decision Making

$\boxed{\text{How much of the data do we really need?}}$

Suppose

$$X_1, \ldots, X_n \sim P_\theta.$$

The complete sample contains information about the unknown parameter θ.

But statistical procedures rarely use the raw data directly.

$$(X_1, \ldots, X_n) \quad \longrightarrow \quad T(X_1, \ldots, X_n) \quad \longrightarrow \quad \text{Inference.}$$

Today we ask: $\boxed{\textbf{When does } T(X) \textbf{ retain all relevant information about } \theta\textbf{?}}$

Learning Objectives

By the end of this lecture, you should be able to

1. Define a statistic and determine its distribution.
2. Explain and characterize sufficiency.
3. Identify minimal sufficient statistics.
4. Define completeness and recognize its role in estimation.
5. Formulate a statistical problem using decision rules, loss, and risk.
6. Explain admissibility and basic notions of optimality.

What Is a Statistic?

Let

$$X = (X_1, \ldots, X_n) \sim P_\theta, \qquad \theta \in \Theta.$$

Definition: Statistic

A **statistic** is a measurable function of the observed data,

$$T = T(X_1, \ldots, X_n),$$

that does not depend on the unknown parameter θ.

Examples include

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad X_{(n)} = \max_i X_i, \qquad \#\text{ of } X_i's \le t = \sum_{i=1}^{n} \mathbf{1}\{X_i \le t\}.$$

A Statistic Is a Random Variable

Before the data are observed,

$$T = T(X_1, \ldots, X_n)$$

is itself a random variable.

Therefore, T has a distribution induced by the distribution of the sample.

$$X \sim P_\theta \quad \Longrightarrow \quad T(X) \sim P_\theta^T.$$

Here

$$P_\theta^T(B) = P_\theta\{T(X) \in B\}$$

is the distribution of T under P_θ.

Statistical inference depends crucially on these sampling distributions.

Example: Distribution of the Sample Mean

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2).$$

Consider the statistic

$$T(X) = \overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Since linear combinations of independent Normal random variables are Normal,

$$\boxed{\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).}$$

Thus the distribution of a statistic generally depends on the unknown parameter.

Another Example: Maximum of Uniform Variables

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Unif}(0, \theta), \qquad \theta > 0.$$

Consider

$$T = X_{(n)} = \max\{X_1, \ldots, X_n\}.$$

For $0 \leq t \leq \theta$,

$$\begin{aligned} P_\theta(T \leq t) &= P_\theta(X_1 \leq t, \ldots, X_n \leq t) \\ &= \prod_{i=1}^{n} P_\theta(X_i \leq t) \\ &= \left(\frac{t}{\theta}\right)^n. \end{aligned}$$

Therefore, $F_T(t) = \left(\frac{t}{\theta}\right)^n, \qquad 0 \leq t \leq \theta.$

Statistics as Data Reduction

A statistic maps the original sample

$$X = (X_1, \ldots, X_n)$$

to a potentially simpler object

$$T(X).$$

For example,

$$(X_1, \ldots, X_n) \quad \longrightarrow \quad \overline{X}.$$

We have compressed n observations into a single number.

Statistics as Data Reduction

$$(X_1, \ldots, X_n) \quad \longrightarrow \quad \overline{X}.$$

But compression raises an important question:

$$\boxed{\text{What information about } \theta \text{ was lost?}}$$

Not every statistic preserves the information contained in the sample.

A Tale of Two Statistics

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

Consider

$$T_1(X) = \sum_{i=1}^{n} X_i$$

and

$$T_2(X) = X_1.$$

Both are statistics. But intuitively,

$$\boxed{T_1 \text{ contains substantially more information about } p \text{ than } T_2.}$$

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Can we make the notion of "retaining all information about p" mathematically precise?

What Should "No Information Loss" Mean?

Suppose we observe a statistic

$$T(X) = t.$$

There may be many samples x that produce the same value t.

If T retains all information about θ, then after knowing $T(X)$, the remaining variation in X should tell us nothing further about θ.

This suggests requiring

$$\boxed{\text{The conditional distribution } \mathcal{L}_\theta(X \mid T) \text{ does not depend on } \theta.}$$

This is the central idea behind **sufficiency**.

Definition of Sufficiency

Let

$$X \sim P_\theta, \qquad \theta \in \Theta,$$

and let $T = T(X)$ be a statistic.

Definition: Sufficient Statistic

The statistic T is **sufficient for** θ if the conditional distribution of X given T does not depend on θ.
Equivalently,

$$\mathcal{L}_\theta(X \mid T = t) \text{ is the same for every } \theta \in \Theta.$$

Once T is known, the remaining randomness in X contains no further information about θ.

Example: Bernoulli Sample

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p).$$

Define

$$T = \sum_{i=1}^{n} X_i.$$

The joint probability of a sample $x = (x_1, \ldots, x_n)$ is

$$P_p(X = x) = p^{\sum_i x_i}(1 - p)^{n - \sum_i x_i}.$$

Hence, for any sample satisfying

$$\sum_i x_i = t \implies P_p(X = x) = p^t(1 - p)^{n-t}.$$

Thus all samples with the same number of successes have the same probability.

Bernoulli Sample: Conditional Distribution

Given

$$T = \sum_{i=1}^{n} X_i = t, \quad \#\{\text{Binary sequence such that } \sum_{i=1}^{n} X_i = t\} = \binom{n}{t}$$

there are $\binom{n}{t}$ possible binary sequences containing exactly t successes.

For each such sequence $(x_1, \ldots, x_n)$,

$$P_p(X_1 = x_1, \ldots, X_n = x_n \mid T = t) = \frac{1}{\binom{n}{t}} \longleftarrow \textbf{does not depend on } p.$$

Therefore,

$$\boxed{T = \sum_{i=1}^{n} X_i \text{ is sufficient for } p.}$$

Sufficiency as Data Compression

For the Bernoulli model,

$$(X_1, \ldots, X_n)$$

contains n observations.

Yet for inference about p, we may replace the entire sample by

$$T = \sum_{i=1}^{n} X_i.$$

Thus

$$\{0,1\}^n \quad \longrightarrow \quad \{0,1,\ldots,n\}.$$

Checking Sufficiency Directly Can Be Difficult

The definition requires studying

$$\mathcal{L}_\theta(X \mid T).$$

For complicated models, computing this conditional distribution may be difficult.

Fortunately, there is a much easier characterization when densities exist.

The Fisher–Neyman Factorization Theorem

Fisher–Neyman Factorization Theorem

Theorem (Factorization Theorem)

Suppose the family $\{P_\theta : \theta \in \Theta\}$ admits densities $f_\theta(x)$ with respect to a common dominating measure.

A statistic $T(X)$ is sufficient for θ iff

$$f_\theta(x) = g_\theta(T(x))\, h(x),$$

where g_θ may depend on θ and $T(x)$, while $h(x)$ does not depend on θ.

The dependence of the likelihood on θ must occur entirely through $T(x)$.

Factorization Theorem: Intuition

Suppose

$$f_\theta(x) = g_\theta(T(x))h(x).$$

Once $T(x)$ is known,

$$g_\theta(T(x))$$

is fixed.

The remaining variation in x occurs only through

$$h(x),$$

which does not involve θ.

Therefore,

Example: Poisson Model

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda).$$

The joint pmf is

$$f_\lambda(x_1, \ldots, x_n) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = e^{-n\lambda}\lambda^{\sum_i x_i} \prod_{i=1}^{n} \frac{1}{x_i!}.$$

Thus

$$g_\lambda(T) = e^{-n\lambda}\lambda^{T},\ T = \sum_{i=1}^{n} X_i, \quad \text{and} \quad h(x) = \prod_{i=1}^{n} \frac{1}{x_i!}.$$

$$\boxed{\sum_{i=1}^{n} X_i \text{ is sufficient for } \lambda.}$$

Example: Normal Model

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$,

with both μ and σ^2 unknown. The likelihood is

$$L(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (X_i - \mu)^2 \right\}.$$

$$\text{Expanding,} \quad \sum_{i=1}^{n} (X_i - \mu)^2 = \sum_{i=1}^{n} X_i^2 - 2\mu \sum_{i=1}^{n} X_i + n\mu^2.$$

Therefore the likelihood depends on the sample only through

$$\boxed{\left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right) \quad \text{is sufficient for } (\mu, \sigma^2).}$$

Exponential Families Revisited

Recall the exponential-family density

$$f_\eta(x) = h(x) \exp\left\{\eta^\top T(x) - A(\eta)\right\}.$$

For an iid sample,

$$L(\eta; x_{1:n}) = \left[\prod_{i=1}^n h(x_i)\right] \exp\left\{\eta^\top \sum_{i=1}^n T(x_i) - nA(\eta)\right\}.$$

By the factorization theorem,

$$\boxed{T_n = \sum_{i=1}^n T(X_i) \text{ is sufficient for } \eta.}$$

.

Why Sufficiency Matters

Sufficiency separates

$$\boxed{\text{information about } \theta}$$

from

$$\boxed{\text{irrelevant sample variation}}.$$

This has several consequences.

- Estimators can often be improved by conditioning on sufficient statistics.
- The dimension of the data may be greatly reduced.
- Likelihood-based inference becomes simpler.
- Exponential families naturally generate sufficient statistics.

Later we will use this idea in the **Rao–Blackwell theorem**.

Is Every Sufficient Statistic Equally Good?

Suppose $T(X)$ is sufficient.

Then so is

$$(T(X), X_1).$$

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Indeed, adding extra information cannot destroy sufficiency.

But this statistic clearly retains more information than necessary.

So we ask a sharper question:

$$\boxed{\text{What is the smallest sufficient summary of the sample?}}$$

This leads to **minimal sufficiency**.

Minimal Sufficiency

A sufficient statistic may still contain unnecessary information.

Definition: Minimal Sufficient Statistic

A sufficient statistic T is **minimal sufficient** if, for every other sufficient statistic S, there exists a measurable function g such that

$$T = g(S) \qquad P_\theta\text{-a.s. for all } \theta.$$

Thus every other sufficient statistic determines T.

Informally,

$$\boxed{T = \text{the coarsest statistic that is still sufficient.}}$$

Minimal Sufficiency as Data Partitioning

Every statistic T partitions the sample space according to

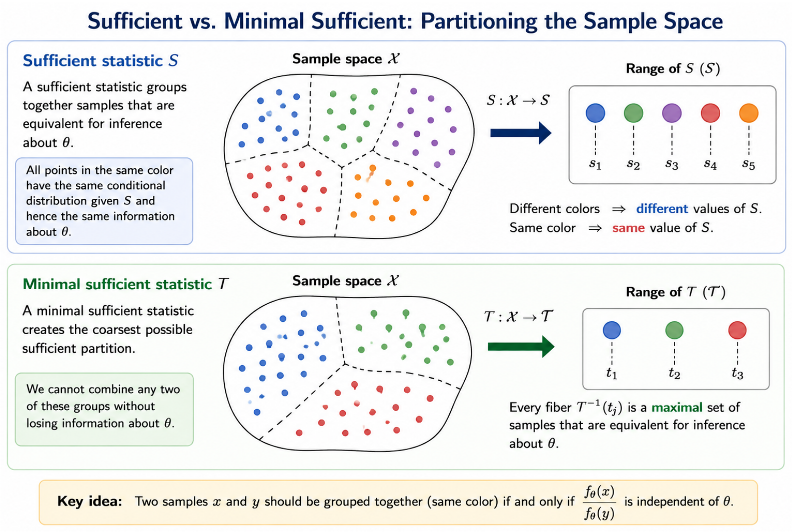
$$x \sim_T y \quad \iff \quad T(x) = T(y).$$

A sufficient statistic groups together samples that are equivalent for inference about θ.

A minimal sufficient statistic creates the **coarsest possible sufficient partition**.

$$\boxed{\text{same value of } T \iff \text{statistically indistinguishable samples}}$$

Minimal Sufficiency as Data Partitioning



A Useful Criterion for Minimal Sufficiency

Suppose the family has densities f_θ with respect to a common dominating measure.

Theorem (Likelihood-Ratio Criterion)

Suppose a statistic T satisfies

$$T(x) = T(y) \quad \Longleftrightarrow \quad \frac{f_\theta(x)}{f_\theta(y)} \text{ is independent of } \theta,$$

whenever the ratio is well-defined.

Then T is minimal sufficient.

This criterion is often the easiest way to identify a minimal sufficient statistic.

Why the Likelihood-Ratio Criterion Makes Sense

Consider two possible samples x and y.

The ratio

$$\frac{f_\theta(x)}{f_\theta(y)}$$

measures their relative plausibility under parameter value θ.

If this ratio does not depend on θ, then observing x rather than y provides no additional information about which θ generated the data.

Hence such samples should be assigned the same value by a minimal sufficient statistic.

Example: Bernoulli Model

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$.

For two samples x and y,

$$\frac{f_p(x)}{f_p(y)} = \frac{p^{\sum_i x_i}(1-p)^{n-\sum_i x_i}}{p^{\sum_i y_i}(1-p)^{n-\sum_i y_i}}.$$

Therefore,

$$\frac{f_p(x)}{f_p(y)} = \left(\frac{p}{1-p}\right)^{\sum_i x_i - \sum_i y_i}.$$

This ratio is independent of p iff

$$\sum_i x_i = \sum_i y_i.$$

Hence, $T(X) = \sum_{i=1}^n X_i$ is minimal sufficient for p.

Example: Normal Model

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, with both parameters unknown.

For two samples x and y,

$$\frac{f_{\mu,\sigma^2}(x)}{f_{\mu,\sigma^2}(y)}$$

is independent of (μ, σ^2) iff

$$\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i \quad \text{and} \quad \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} y_i^2.$$

$$\boxed{T(X) = \left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right) \text{ is minimal sufficient for } (\mu, \sigma^2).}$$

Sufficient versus Minimal Sufficient

Sufficient	Minimal Sufficient
Retains all information about θ	Retains all information with maximum compression
May contain extra information	Coarsest sufficient summary
Often found by factorization	Often found by likelihood ratios

Sufficiency asks

"Have we lost information?"

Minimal sufficiency asks

"Have we retained more than necessary?"

Which Estimator Should We Use?

Suppose our goal is to estimate a quantity $\tau(\theta)$. There may be many unbiased estimators:

$$\delta_1(X), \quad \delta_2(X), \quad \delta_3(X), \quad \ldots$$

all satisfying

$$\mathbb{E}_\theta[\delta_j(X)] = \tau(\theta).$$

How should we choose among them?

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How should we choose among them?

A natural criterion is variance:

$$\boxed{\text{Among unbiased estimators, find one with the smallest variance.}}$$

Such an estimator is called a **uniformly minimum-variance unbiased (UMVU) estimator**.

Can Sufficiency Help Us Find the Best Estimator?

Suppose T is sufficient for θ.

Given any unbiased estimator $\delta(X)$, conditioning on T gives

$$\delta^*(T) = \mathbb{E}_\theta[\delta(X) \mid T].$$

Under suitable conditions, $\delta^*(T)$ remains unbiased and satisfies

$$\mathrm{Var}_\theta\{\delta^*(T)\} \leq \mathrm{Var}_\theta\{\delta(X)\}.$$

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This is the idea behind the **Rao–Blackwell theorem**.

Thus, when searching for the best unbiased estimator, it is enough to look among functions of a sufficient statistic.

Rao–Blackwell Theorem

Suppose $\delta(X)$ is an unbiased estimator of $\tau(\theta)$, and $T = T(X)$ is sufficient for θ.

Define the Rao–Blackwellized estimator

$$\delta^*(T) = \mathbb{E}_\theta[\delta(X) \mid T].$$

Since T is sufficient, the conditional distribution of X given T does not depend on θ, so $\delta^*(T)$ is a statistic.

Theorem (Rao–Blackwell)

The estimator $\delta^*(T)$ is unbiased for $\tau(\theta)$ and

$$\mathrm{Var}_\theta\{\delta^*(T)\} \leq \mathrm{Var}_\theta\{\delta(X)\}$$

for every $\theta \in \Theta$.

Why Does Rao–Blackwell Work?

By the law of total variance,

$$\mathrm{Var}_\theta\{\delta(X)\} = \mathrm{Var}_\theta\left\{\mathbb{E}_\theta[\delta(X) \mid T]\right\} + \mathbb{E}_\theta\left[\mathrm{Var}_\theta\{\delta(X) \mid T\}\right].$$

Since the second term is nonnegative,

$$\mathrm{Var}_\theta\{\delta(X)\} \geq \mathrm{Var}_\theta\{\delta^*(T)\}.$$

Also, by the tower property,

$$\mathbb{E}_\theta[\delta^*(T)] = \mathbb{E}_\theta[\delta(X)] = \tau(\theta).$$

Therefore conditioning on a sufficient statistic can only improve an unbiased estimator in terms of variance.

$\boxed{\text{To search for the best unbiased estimator, look among functions of } T.}$

But Which Function of T?

Suppose T is sufficient.

Rao–Blackwell tells us to search among estimators of the form

$$g(T).$$

But there may still be many functions of T satisfying

$$\mathbb{E}_\theta[g(T)] = \tau(\theta) \qquad \text{for every } \theta.$$

How do we know when we have found the optimal unbiased estimator?

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How do we know when we have found the optimal unbiased estimator?

We need an additional property of the sufficient statistic.

Completeness

Completeness

Sufficiency concerns whether a statistic retains all information about the parameter.

Completeness concerns whether that information is represented **without hidden redundancies**.

$$\boxed{\text{Sufficiency: information preservation}}$$

$$\boxed{\text{Completeness: uniqueness of unbiased information}}$$

We now make this precise.

Definition of Completeness

Let T be a statistic whose distribution depends on θ.

Definition: Complete Statistic

The statistic T is **complete** for $\{P_\theta : \theta \in \Theta\}$ if, for every measurable function g,

$$\mathbb{E}_\theta[g(T)] = 0 \qquad \text{for all } \theta \in \Theta$$

implies

$$P_\theta\{g(T) = 0\} = 1 \qquad \text{for all } \theta \in \Theta.$$

Thus, there is no nontrivial function of T whose expectation vanishes for every value of the parameter.

What Does Completeness Mean?

Suppose T is complete and two functions $g_1(T)$ and $g_2(T)$ satisfy

$$\mathbb{E}_\theta[g_1(T)] = \mathbb{E}_\theta[g_2(T)] \qquad \text{for every } \theta.$$

Then

$$\mathbb{E}_\theta\left[g_1(T) - g_2(T)\right] = 0 \qquad \text{for every } \theta.$$

By completeness,

$$g_1(T) = g_2(T) \qquad \text{a.s.}$$

Therefore,

$$\boxed{\text{unbiased functions of a complete statistic are unique.}}$$

Example: Bernoulli Statistic

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$, and let

$$T = \sum_{i=1}^{n} X_i \sim \text{Binomial}(n, p).$$

Suppose

$$\mathbb{E}_p[g(T)] = 0 \qquad \text{for every } p \in (0, 1).$$

Then

$$\sum_{t=0}^{n} g(t) \binom{n}{t} p^t (1-p)^{n-t} = 0 \qquad \text{for every } p \in (0, 1). \tag{1}$$

Bernoulli Statistic Is Complete

divide (1) by $(1-p)^n$:

$$\sum_{t=0}^{n} g(t)\binom{n}{t}\left(\frac{p}{1-p}\right)^t = 0.$$

substituting $z = \frac{p}{1-p}$ yields

$$\sum_{t=0}^{n} g(t)\binom{n}{t} z^t = 0 \qquad \text{for every } z > 0.$$

A polynomial that vanishes on an interval must have every coefficient equal to zero.

$$g(t) = 0, \qquad t = 0, \ldots, n. \Longrightarrow g(T) \overset{a.s.}{=} 0.$$

Therefore $T = \sum_i X_i$ is complete.

Completeness in Exponential Families

Recall a k-parameter exponential family $f_\eta(x) = h(x) \exp\left\{\eta^\top T(x) - A(\eta)\right\}$.

For an iid sample, the natural statistic is

$$T_n = \sum_{i=1}^{n} T(X_i).$$

Under suitable regularity conditions, if the natural parameter space contains an open subset of $\mathbb{R}^k$, then T_n is complete.

Remark: Important

Sufficiency alone does **not** imply completeness. Completeness requires additional structure on the family.

Complete and Sufficient: Why Do We Care?

Suppose T is both

sufficient and **complete**.

Sufficiency says that T retains all information about θ.

Completeness says that unbiased functions of T are uniquely determined.

Together, these properties give a remarkably strong result.

$$\boxed{\text{Complete} + \text{Sufficient} \quad \Longrightarrow \quad \text{Optimal Unbiased Estimation}}$$

This is the content of the **Lehmann–Scheffé theorem**.

Lehmann–Scheffé Theorem

Theorem (Lehmann–Scheffé)

Suppose T is a **complete sufficient** *statistic for θ.*

If $\delta(T)$ is unbiased for a parameter $\tau(\theta)$, i.e.,

$$\mathbb{E}_\theta[\delta(T)] = \tau(\theta) \qquad \text{for all } \theta,$$

then $\delta(T)$ is the unique uniformly minimum-variance unbiased estimator of $\tau(\theta)$.

Thus completeness converts an unbiased estimator based on T into an optimal estimator within the class of unbiased estimators.

Example: Estimating a Bernoulli Probability

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$.

We have shown that

$$T = \sum_{i=1}^{n} X_i$$

is complete and sufficient for p.

Consider

$$\widehat{p} = \frac{T}{n} = \overline{X} \quad \text{and} \quad \mathbb{E}_p[\widehat{p}] = \frac{1}{n}\mathbb{E}_p[T] = p,$$

The Lehmann–Scheffé theorem implies

$$\boxed{\overline{X} \text{ is the unique UMVU estimator of } p.}$$

The Hierarchy So Far

We started with the complete sample

$$X = (X_1, \ldots, X_n).$$

Then we asked successively:

Statistic	:	How can we summarize the data?
Sufficient	:	Did we retain all information?
Minimal sufficient	:	Did we remove all redundancy?
Complete	:	Are unbiased functions uniquely identified?

These concepts describe the **information contained in data**.

A different question: **How should we use that information to make decisions?**

From Inference to Decisions

So far, our goal has been to learn about an unknown parameter

$$\theta \in \Theta.$$

But statistical inference ultimately requires us to **act** based on the observed data.

For example:

- estimate an unknown parameter,
- accept or reject a scientific hypothesis,
- classify a patient as diseased or healthy,
- choose among competing treatments.

These problems can all be described within a common framework:

$$\boxed{\text{Statistical Decision Theory}}$$

A Statistical Decision Problem

A statistical decision problem consists of four ingredients.

Object	Meaning
Θ	Parameter space
$\mathcal{X}$	Sample space
$\mathcal{A}$	Action space
$L(\theta, a)$	Loss incurred by taking action a

The data are generated according to

$$X \sim P_\theta, \qquad \theta \in \Theta.$$

After observing X, we must choose an action $a \in \mathcal{A}$.

Decision Rules

How should the observed data determine our action?

Definition: Decision Rule

A **decision rule** is a measurable function

$$\delta : \mathcal{X} \to \mathcal{A}.$$

After observing $X = x$, we take the action

$$a = \delta(x).$$

Thus the basic statistical pipeline is

$$\boxed{\theta \longrightarrow X \sim P_\theta \longrightarrow \delta(X) \longrightarrow \text{action}}$$

Estimation as a Decision Problem

Suppose we want to estimate

$$\tau(\theta) \in \mathbb{R}.$$

Take the action space to be

$$\mathcal{A} = \mathbb{R}.$$

A decision rule is simply an estimator

$$\delta(X) = \widehat{\tau}.$$

For example, under squared-error loss,

$$L(\theta, a) = (a - \tau(\theta))^2 .$$

Thus ordinary point estimation is a special case of statistical decision theory.

Hypothesis Testing as a Decision Problem

Suppose we wish to test

$$H_0 : \theta \in \Theta_0 \qquad \text{versus} \qquad H_1 : \theta \in \Theta_1.$$

Take

$$\mathcal{A} = \{0, 1\},$$

where

$$a = 0 \quad \text{means "do not reject } H_0\text{"}, \qquad a = 1 \quad \text{means "reject } H_0\text{"}.$$

$$\delta : \mathcal{X} \to \{0, 1\}.$$

Estimation and testing are instances of the same framework.

Not All Mistakes Are Equally Costly

Suppose an AI system predicts whether a medical scan indicates disease.

There are two possible mistakes:

Truth	Decision	Consequence
Healthy	Disease	False positive
Disease	Healthy	False negative

These errors need not have the same consequence.

Not All Mistakes Are Equally Costly

Suppose an AI system predicts whether a medical scan indicates disease.

There are two possible mistakes:

Truth	Decision	Consequence
Healthy	Disease	False positive
Disease	Healthy	False negative

These errors need not have the same consequence.

Statistical decision theory represents this explicitly through a **loss function**.

$$\boxed{L(\theta, a) = \text{cost of taking action } a \text{ when the truth is } \theta.}$$

Loss Functions

Definition: Loss Function

A loss function

$$L : \Theta \times \mathcal{A} \to [0, \infty)$$

assigns a cost to every combination of the true parameter θ and action a.

For estimation, common choices include

$$L(\theta, a) = (a - \theta)^2 \qquad \text{(squared-error loss)},$$

and

$$L(\theta, a) = |a - \theta| \qquad \text{(absolute-error loss)}.$$

Different loss functions can lead to different optimal procedures.

Loss versus Risk

The loss

$$L(\theta, \delta(X))$$

is random because the decision $\delta(X)$ depends on the random data X.

Before observing the sample, we evaluate a decision rule by its **expected loss**.

Definition: Risk Function

The risk of a decision rule δ at parameter value θ is

$$R(\theta, \delta) = \mathbb{E}_\theta \left[L(\theta, \delta(X)) \right].$$

Thus

$$\boxed{\text{Risk} = \text{expected loss under repeated sampling.}}$$

Example: Risk under Squared-Error Loss

Suppose $\delta(X)$ estimates θ and

$$L(\theta, \delta) = (\delta - \theta)^2.$$

Then

$$R(\theta, \delta) = \mathbb{E}_\theta\left[(\delta(X) - \theta)^2\right].$$

Hence the risk is exactly the mean squared error:

$$\boxed{R(\theta, \delta) = \mathrm{MSE}_\theta(\delta).}$$

Using the bias–variance decomposition,

$$R(\theta, \delta) = \mathrm{Var}_\theta(\delta) + \mathrm{Bias}_\theta(\delta)^2.$$

Decision theory allows us to compare biased and unbiased estimators within a common framework.

How Do We Compare Decision Rules?

Suppose δ_1 and δ_2 are two decision rules.

Each has a risk function

$$R(\theta, \delta_1), \qquad R(\theta, \delta_2).$$

Ideally, one rule would have smaller risk for every possible value of θ.

$$R(\theta, \delta_1) \leq R(\theta, \delta_2) \qquad \text{for all } \theta.$$

If the inequality is strict somewhere, there is no reason to prefer δ_2.

This leads to the notion of **domination**.

Domination

Definition: Domination

A decision rule δ_1 **dominates** δ_2 if

$$R(\theta, \delta_1) \leq R(\theta, \delta_2) \qquad \text{for every } \theta \in \Theta,$$

and

$$R(\theta, \delta_1) < R(\theta, \delta_2)$$

for at least one $\theta \in \Theta$.

A dominated decision rule is uniformly worse than another available procedure.

Admissibility

Definition: Admissible Decision Rule

A decision rule δ is **admissible** iff no other decision rule δ' dominates δ, i.e.,

$$\nexists\theta := \theta(\delta') \quad \text{such that} \quad R(\theta, \delta') > R(\theta, \delta) \quad \textbf{OR} \quad R(\theta, \delta') = R(\theta, \delta) \quad \forall\, \theta \in \Theta.$$

If such a δ' exists, then δ is **inadmissible**.

$$\boxed{\text{Admissible} \quad \Longleftrightarrow \quad \text{not uniformly improvable.}}$$

Example: Bernoulli

Suppose

$$X \sim \text{Bernoulli}(\theta) \quad \text{where} \quad \theta \in [0, 1]$$

,

and consider the squared-error loss.

Consider

$$\delta(X) = X \quad (\text{Is } \delta \text{ admissible?})$$

Example: Bernoulli

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Consider

$$\delta(X) = X \quad (\text{Is } \delta \text{ admissible?})$$

If not, then there exists δ' that dominates δ.

Case 1: If $\theta = 0$, then $X = 0$ w.p. 1. $\mathbb{E}(\delta'(X) - 0)^2 \leq \mathbb{E}(\delta(X) - 0)^2 = 0 \implies \delta'(0) = 0$.

Case 2: If $\theta = 1$, then $X = 1$ w.p. 1. $\mathbb{E}(\delta'(X) - 1)^2 \leq \mathbb{E}(\delta(X) - 1)^2 = 0 \implies \delta'(1) = 1$.
$\boldsymbol{\delta'(X) = X = \delta(X)}$.

Example: Two Estimators of a Normal Mean

Suppose

$$X \sim N(\theta, 1),$$

and consider squared-error loss.

Compare

$$\delta_1(X) = X$$

with the shrinkage estimator

$$\delta_a(X) = aX, \qquad 0 \leq a \leq 1.$$

For δ_1, $R(\theta, \delta_1) = \mathbb{E}_\theta[(X - \theta)^2] = 1$.

What is the risk of δ_a?

Risk of the Shrinkage Estimator

Since

$$aX - \theta = a(X - \theta) - (1 - a)\theta,$$

we obtain

$$R(\theta, \delta_a) = a^2 + (1 - a)^2\theta^2.$$

Thus

$$\boxed{R(\theta, \delta_a) = a^2 + (1 - a)^2\theta^2}$$

whereas $R(\theta, \delta_1) = 1$.

For θ near zero, shrinkage can have smaller risk.

For sufficiently large $|\theta|$, it has larger risk.

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For θ near zero, shrinkage can have smaller risk.

For sufficiently large $|\theta|$, it has larger risk.

Neither estimator dominates the other.

Admissible Does Not Mean Best

Admissibility is a relatively weak optimality requirement.

It says only:

$$\boxed{\text{No other procedure is uniformly better.}}$$

It does **not** say that the procedure has the smallest risk for every θ.

Indeed, two admissible procedures may have crossing risk curves:

$$R(\theta, \delta_1) < R(\theta, \delta_2)$$

for some θ, while

$$R(\theta, \delta_1) > R(\theta, \delta_2) \quad \text{for others.}$$

So how should we choose among admissible procedures?

What Does “Optimal” Mean?

Because risk depends on θ,

$$R(\theta, \delta),$$

there may be no single rule minimizing risk simultaneously for every θ.

Different principles therefore lead to different notions of optimality.

Principle	Objective
Pointwise	Minimize $R(\theta, \delta)$ for each θ
Minimax	Minimize worst-case risk
Bayes	Minimize average risk under a prior

We will study Bayes and minimax procedures in detail later in the course.

The Minimax Principle

A conservative approach is to evaluate a rule by its worst-case risk:

$$\sup_{\theta \in \Theta} R(\theta, \delta).$$

A **minimax rule** solves

$$\boxed{\delta^* \in \arg\min_{\delta} \sup_{\theta \in \Theta} R(\theta, \delta).}$$

The principle is:

Choose the rule whose worst possible performance is best.

This requires no probability distribution over θ.

The Bayes Principle

Alternatively, suppose we assign a prior distribution

$$\theta \sim \pi.$$

The **Bayes risk** of δ is the average risk

$$r(\pi, \delta) = \int_\Theta R(\theta, \delta)\, \pi(d\theta).$$

A Bayes rule satisfies

$$\boxed{\delta_\pi \in \arg\min_\delta r(\pi, \delta).}$$

Thus:

$$\text{Minimax} \quad \longleftrightarrow \quad \text{worst-case performance,}$$

$$\text{Bayes} \quad \longleftrightarrow \quad \text{average performance under } \pi.$$

Lecture 2: The Big Picture

Today we followed two fundamental questions.

1. How much information in the data do we need?

$$\text{Statistic} \to \text{Sufficiency} \to \text{Minimal Sufficiency} \to \text{Completeness}.$$

2. How should we use that information?

$$\text{Decision Rule} \to \text{Loss} \to \text{Risk} \to \text{Admissibility} \to \text{Optimality}.$$

Together, these form two central pillars of mathematical statistics:

$$\boxed{\text{Information} \qquad + \qquad \text{Decision Making}}$$

Next up

We now have the language needed to ask:

$\boxed{\text{How should we construct good estimators from data?}}$

Next: Point Estimation and Likelihood Methods